

P8109 Lecture 11 — Problem Solving Session: Solutions

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Example 1

Problem. Show that the first order statistic $X_{(1)}$ of a random sample of size n from the distribution with probability density function

$$f(x | \theta) = e^{-(x-\theta)}, \quad \theta < x < \infty, \quad -\infty < \theta < \infty,$$

and $f(x | \theta) = 0$ elsewhere, is a complete sufficient statistic for θ . Find the unique function of this statistic which is the minimum variance unbiased estimator (MVUE) of θ .

Solution.

Sufficiency. $L(\theta | \mathbf{x}) = e^{-\sum x_i + n\theta} \mathbf{1}_{\{x_{(1)} > \theta\}}$. By factorization (Lect. 6), $T = X_{(1)}$ is sufficient.

Distribution of T . $P(T > t) = e^{-n(t-\theta)}$ for $t > \theta$, so $T - \theta \sim \text{Exp}(n)$ with density $f_T(t) = ne^{-n(t-\theta)} \mathbf{1}_{\{t > \theta\}}$.

Completeness. Suppose $\mathbb{E}_\theta h(T) = 0$ for all $\theta \in \mathbb{R}$:

$$\int_\theta^\infty h(t) ne^{-n(t-\theta)} dt = 0 \iff \int_\theta^\infty h(t) e^{-nt} dt = 0 \quad \forall \theta.$$

Differentiate w.r.t. θ (FTC, as in Ex. 6.5): $-h(\theta)e^{-n\theta} = 0 \Rightarrow h \equiv 0$. Hence T is complete.

MVUE of θ . $\mathbb{E}_\theta T = \theta + \frac{1}{n}$. By Lehmann-Scheffé (Thm 6.3), the unique MVUE is

$$\hat{\theta} = X_{(1)} - \frac{1}{n}.$$

Example 2

Problem. Let a random sample of size n be taken from a discrete distribution with probability mass function

$$f(x | \theta) = \frac{1}{\theta}, \quad x = 1, 2, \dots, \theta,$$

and $f(x | \theta) = 0$ elsewhere, where θ is an unknown positive integer. **(a)** Show that the largest observation, say $T = X_{(n)}$, of the sample is a complete sufficient statistic for θ . **(b)** Prove that

$$\frac{T^{n+1} - (T-1)^{n+1}}{T^n - (T-1)^n}$$

is the unique minimum variance unbiased estimator (MVUE) of θ .

Solution.

(a) $L(\theta | \mathbf{x}) = \theta^{-n} \mathbf{1}_{\{x_{(n)} \leq \theta\}} \mathbf{1}_{\{x_{(1)} \geq 1\}}$, so $T = X_{(n)}$ is sufficient.

Distribution: $P(T \leq t) = (t/\theta)^n$, so $P(T = t) = [t^n - (t-1)^n]/\theta^n$ for $t = 1, \dots, \theta$.

Completeness: if $\mathbb{E}_\theta h(T) = \sum_{t=1}^\theta h(t) [t^n - (t-1)^n]/\theta^n = 0$ for every $\theta \in \mathbb{Z}_{>0}$, then

$$\sum_{t=1}^\theta h(t) [t^n - (t-1)^n] = 0 \quad \forall \theta.$$

Take $\theta = 1$: $h(1) = 0$. Subtract consecutive θ 's: $h(\theta)[\theta^n - (\theta-1)^n] = 0 \Rightarrow h(\theta) = 0$. By induction, $h \equiv 0$, so T is complete.

(b) Need $g(T)$ with $\mathbb{E}_\theta g(T) = \theta$, i.e.,

$$\sum_{t=1}^\theta g(t) [t^n - (t-1)^n] = \theta^{n+1} \quad \forall \theta.$$

Subtracting the equation at $\theta - 1$ from that at θ :

$$g(\theta) [\theta^n - (\theta-1)^n] = \theta^{n+1} - (\theta-1)^{n+1}.$$

Therefore the unique MVUE (Lehmann–Scheffé) is

$$g(T) = \frac{T^{n+1} - (T-1)^{n+1}}{T^n - (T-1)^n}.$$

Example 3

Problem. Let $X_1, \dots, X_n$ be a random sample of size n from a distribution with probability density function

$$f(x | \theta) = \frac{1}{2} e^{-|x-\theta|}, \quad -\infty < x < \infty,$$

for all real θ . Find the likelihood ratio test Λ for testing

$$H_0 : \theta = 0 \quad \text{against} \quad H_1 : \theta \neq 0.$$

Solution. The MLE of θ for the double exponential is the sample median $\widetilde{M}$ (the value of c minimizing $\sum |x_i - c|$). Hence

$$\Lambda = \frac{L(0)}{L(\widetilde{M})} = \exp \left(\sum_{i=1}^n |X_i - \widetilde{M}| - \sum_{i=1}^n |X_i| \right) \leq 1.$$

Reject H_0 when $\Lambda \leq \lambda_0$, equivalently when $\sum |X_i| - \sum |X_i - \widetilde{M}|$ is large. (By Thm 9.1, large-sample test rejects when $-2 \log \Lambda > \chi_{1,\alpha}^2$.)

Example 4

Problem. Consider a random sample $X_1, X_2, \dots, X_n$ from a distribution with probability density function

$$f(x | \theta) = \theta(1 - x)^{\theta-1}, \quad 0 < x < 1,$$

and $f(x | \theta) = 0$ elsewhere, where $\theta > 0$. **(a)** Find the form of the uniformly most powerful test of $H_0 : \theta = 1$ against $H_1 : \theta > 1$. **(b)** What is the likelihood ratio Λ for testing $H_0 : \theta = 1$ against $H_1 : \theta \neq 1$?

Solution. Let $T = -\sum_{i=1}^n \log(1 - X_i)$. Under $\theta = 1$, $-\log(1 - X_i) \sim \text{Exp}(1)$, so $2T \sim \chi_{2n}^2$.

(a) $L(\theta) = \theta^n \exp\{-(\theta - 1)T\}$. For $\theta_1 > \theta_0$,

$$\frac{L(\theta_1)}{L(\theta_0)} = \left(\frac{\theta_1}{\theta_0}\right)^n e^{-(\theta_1 - \theta_0)T},$$

which is *decreasing* in T , i.e., MLR in $-T$. By Thm 8.1, the UMP test of $H_0 : \theta = 1$ vs $H_1 : \theta > 1$ rejects for **small** T :

Reject H_0 iff $2T < \chi_{2n, 1-\alpha}^2$

where $\chi_{2n, 1-\alpha}^2$ is the lower α -quantile of χ_{2n}^2 .

(b) $L(1) = 1$. Score equation gives MLE $\hat{\theta} = n/T$, so

$$\sup_{\theta} L(\theta) = (n/T)^n e^{T-n}.$$

Therefore

$\Lambda = (T/n)^n e^{n-T}$

reject H_0 for small Λ .

Example 5

Problem. Let $X_1, X_2, \dots, X_n$ denote a random sample from a distribution that is $N(0, \theta)$, where the variance θ is an unknown positive number. Does there exist a uniformly most powerful test with significance level α for testing the simple hypothesis

$$H_0 : \theta = 1$$

against the composite alternative hypothesis $H_1 : \theta > 1$?

Solution. $L(\theta) = (2\pi\theta)^{-n/2} \exp(-\frac{1}{2\theta} \sum X_i^2)$. This is a one-parameter exponential family with sufficient statistic $T = \sum X_i^2$ and natural parameter $\eta(\theta) = -1/(2\theta)$, which is *strictly increasing* in θ . Hence MLR in T (Ex. 8.2 form).

By Thm 8.1, **yes**, the UMP size- α test exists:

Reject H_0 iff $\sum X_i^2 > \chi_{n, \alpha}^2$

since under $\theta = 1$, $\sum X_i^2 \sim \chi_n^2$.

Example 6

Problem. Let $X_1, \dots, X_n$ be i.i.d. from a discrete distribution with probability mass function

$$f(x | p) = p(1 - p)^x, \quad x = 0, 1, 2, \dots$$

Derive the uniformly most powerful (UMP) test of size α for

$$H_0 : p \leq 0.5 \quad \text{versus} \quad H_1 : p > 0.5.$$

Solution. $L(p) = p^n(1 - p)^S$, $S = \sum X_i$. For $p_1 > p_0$,

$$\frac{L(p_1)}{L(p_0)} = \left(\frac{p_1}{p_0}\right)^n \left(\frac{1-p_1}{1-p_0}\right)^S,$$

decreasing in S (since $(1 - p_1)/(1 - p_0) < 1$). Thus MLR in $-S$. By Thm 8.1, the UMP test rejects for **small** S :

$$\phi(\mathbf{X}) = \begin{cases} 1, & S < c_\alpha, \\ \gamma_\alpha, & S = c_\alpha, \\ 0, & S > c_\alpha, \end{cases}$$

where $(c_\alpha, \gamma_\alpha)$ are determined by $\mathbb{E}_{p=1/2}\phi = \alpha$, and $S \sim \text{NegBin}(n, p)$ with $P(S = k) = \binom{n+k-1}{k} p^n (1-p)^k$.

Example 7

Problem. Let $X_1, X_2, \dots, X_n$ denote a random sample from a Poisson distribution with mean θ , where $0 < \theta < \infty$. Let Θ be an observed value of the random variable Θ . If Θ has the prior probability density function

$$h(\theta) = \frac{\theta^{\alpha-1} e^{-\theta/\beta}}{\Gamma(\alpha) \beta^\alpha}, \quad 0 < \theta < \infty,$$

and $h(\theta) = 0$ elsewhere, where $\alpha > 0$ and $\beta > 0$ are known constants. Find the Bayes estimator of θ under the squared error loss.

Solution. Likelihood $\times$ prior:

$$\pi(\theta | \mathbf{x}) \propto \theta^{\sum x_i} e^{-n\theta} \cdot \theta^{\alpha-1} e^{-\theta/\beta} = \theta^{\alpha + \sum x_i - 1} \exp\left(-\theta\left[n + \frac{1}{\beta}\right]\right).$$

Hence (Lect. 10)

$$\theta | \mathbf{X} \sim \text{Gamma}\left(\alpha + \sum X_i, \frac{\beta}{1 + n\beta}\right).$$

Under squared error loss, the Bayes estimator is the posterior mean (Cor. 10.1):

$$\boxed{\hat{\theta}_B = \frac{\beta(\alpha + \sum_{i=1}^n X_i)}{1 + n\beta}}.$$

Example 8

Problem. Consider a Bayesian model in which θ is absolutely continuous with density

$$\lambda(\theta) = \frac{4\theta^3}{(1+\theta)^5}, \quad \theta > 0.$$

Given $\theta > 0$, the data X and Y have joint density

$$f(x, y | \theta) = \frac{1}{2\theta^3}, \quad |x| < \theta, \quad 0 < y < \theta^2.$$

Find the Bayes estimate of θ under the squared error loss.

Solution. Joint posterior kernel:

$$\lambda(\theta)f(x, y | \theta) = \frac{4\theta^3}{(1+\theta)^5} \cdot \frac{1}{2\theta^3} \mathbf{1}_{\{|x| < \theta, 0 < y < \theta^2\}} = \frac{2}{(1+\theta)^5} \mathbf{1}_{\{\theta > m\}},$$

where $m = \max(|x|, \sqrt{y})$. Normalize:

$$\int_m^\infty \frac{2}{(1+\theta)^5} d\theta = \frac{1}{2(1+m)^4} \implies \pi(\theta | x, y) = \frac{4(1+m)^4}{(1+\theta)^5}, \quad \theta > m.$$

With $u = 1 + \theta$:

$$\mathbb{E}[\theta | x, y] = 4(1+m)^4 \int_{1+m}^\infty \frac{u-1}{u^5} du = 4(1+m)^4 \left[\frac{(1+m)^{-3}}{3} - \frac{(1+m)^{-4}}{4} \right] = \frac{4(1+m)}{3} - 1.$$

Bayes estimate under squared-error loss (posterior mean):

$$\boxed{\hat{\theta}_B = \frac{4 \max(|X|, \sqrt{Y}) + 1}{3}}.$$

Final 22 Q2(b)

Problem. Let $X \sim N(0, \theta)$, $\theta > 0$. **(a)** Show that X is not complete. **(b)** Show that X^2 is complete.

Solution.

(a) $\mathbb{E}_\theta[X] = 0$ for all $\theta > 0$, but $h(x) = x \not\equiv 0$. So X is not complete.

(b) The density $f(x | \theta) = \frac{1}{\sqrt{2\pi\theta}} \exp(-\frac{x^2}{2\theta}) = h(x) \exp\{\eta T(x) - A(\eta)\}$ with $T(x) = x^2$ and $\eta = -1/(2\theta)$ ranging over $(-\infty, 0)$, an open interval. This is a one-parameter exponential family of full rank, so by Thm 6.2, $T(X) = X^2$ is complete.

(Direct: setting $s = 1/(2\theta)$, $\mathbb{E}h(X^2) = 0$ becomes $\int_0^\infty h(y)y^{-1/2}e^{-sy}dy = 0$ for all $s > 0$. Uniqueness of the Laplace transform forces $h \equiv 0$.)

Final 22 Q3

Problem. Let $X_1, \dots, X_n$ be i.i.d. random variables, each with a $\text{Poisson}(\theta)$ distribution. **(a)** Show that $S = \sum_{i=1}^n X_i$ is a sufficient statistic for θ . **(b)** Show that the statistic

$$T(X) = \begin{cases} 1, & \text{if } X = k, \\ 0, & \text{otherwise,} \end{cases}$$

is unbiased for $g(\theta) = P(X = k)$. **(c)** Use the Rao–Blackwell theorem to show that

$$\binom{S}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{S-k}$$

is an improved unbiased estimator of $g(\theta)$.

Solution.

(a) $f(\mathbf{x} \mid \theta) = \frac{e^{-n\theta} \theta^{\sum x_i}}{\prod x_i!}$. By factorization, $S = \sum X_i$ is sufficient.

(b) $\mathbb{E}[\mathbf{1}_{\{X_1=k\}}] = P(X_1 = k) = g(\theta)$.

(c) Conditional on $S = s$, $X_1 \sim \text{Bin}(s, 1/n)$ (standard Poisson splitting). Hence by Rao–Blackwell:

$$\mathbb{E}[T(X_1) \mid S] = P(X_1 = k \mid S) = \binom{S}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{S-k}.$$

This is unbiased for $g(\theta)$ and has variance no larger than that of $T(X_1)$.

Final 22 Q4

Problem. Let X be a random variable from two possible distributions. Under H_0 , $X \sim f_0(x)$, where

$$f_0(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad -\infty < x < \infty.$$

Under H_1 , $X \sim f_1(x)$, where

$$f_1(x) = \frac{1}{2} e^{-|x|}, \quad -\infty < x < \infty.$$

Based on a single observation, derive the most powerful (MP) size- α test.

Solution. By NP Lemma, MP test rejects when $f_0(x)/f_1(x) < k$, i.e.,

$$\frac{(2\pi)^{-1/2} e^{-x^2/2}}{\frac{1}{2} e^{-|x|}} = \sqrt{\frac{2}{\pi}} \exp\left(|x| - \frac{x^2}{2}\right) < k.$$

Equivalently $|x| - x^2/2 < c$, i.e., $(|x| - 1)^2 > 1 - 2c$. Letting $d = \sqrt{1 - 2c}$, the rejection region is

$$\{|X| < 1 - d\} \cup \{|X| > 1 + d\} \quad (0 \leq d \leq 1).$$

Choose d so the size equals α . Under H_0 , $X \sim N(0, 1)$:

$$2\Phi(1 - d) - 1 + 2[1 - \Phi(1 + d)] = \alpha \iff \Phi(1 + d) - \Phi(1 - d) = \frac{1 - \alpha}{2}.$$

The unique $d \in [0, 1]$ satisfying this defines the MP size- α test. (For very small $\alpha < 2(1 - \Phi(2)) \approx 0.046$ the inner region collapses and the test reduces to $|X| > z_{\alpha/2}$.)

Final 22 Q5

Problem. A statistician observes a random variable X having a uniform distribution on $(0, \theta)$, where θ is assumed to have the prior

$$g(\theta) = \begin{cases} \theta e^{-\theta}, & \text{if } \theta > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

The aim is to estimate θ with quadratic loss function $L(\theta, T) = (T - \theta)^2$, where T is an estimator of θ . **(a)** By writing the posterior distribution of θ explicitly, or otherwise, obtain the posterior mean of θ . **(b)** Obtain the posterior expected loss, and by minimizing it, calculate the Bayes estimator. **(c)** Comment on your answers in (a) and (b).

Solution.

(a) Joint density:

$$f(x, \theta) = \frac{1}{\theta} \cdot \theta e^{-\theta} \mathbf{1}_{\{0 < x < \theta\}} = e^{-\theta} \mathbf{1}_{\{\theta > x\}}.$$

Marginal of X : $\int_x^\infty e^{-\theta} d\theta = e^{-x}$. Hence posterior

$$\pi(\theta | x) = \frac{e^{-\theta}}{e^{-x}} \mathbf{1}_{\{\theta > x\}} = e^{-(\theta - x)} \mathbf{1}_{\{\theta > x\}},$$

i.e., $\theta | x \sim x + \text{Exp}(1)$. Posterior mean:

$$\mathbb{E}[\theta | x] = x + 1.$$

(b) Posterior expected loss for an action T :

$$\mathbb{E}[(T - \theta)^2 | x] = T^2 - 2T \mathbb{E}[\theta | x] + \mathbb{E}[\theta^2 | x],$$

minimized at $T = \mathbb{E}[\theta | x] = x + 1$. So the Bayes estimator is

$$\boxed{\hat{\theta}_B = X + 1}.$$

(c) Both answers agree, illustrating Thm 10.1 / Cor. 10.1: under squared-error loss, the Bayes estimator is the posterior mean.

Final 24 Q3

Problem. Consider a random sample $X_1, X_2, \dots, X_n$ from a $N(\mu, \mu^2)$ distribution. **(a)** By using the Fisher–Neyman factorization theorem, find a sufficient statistic in the form (T_1, T_2) for μ . **(b)** Show that

$$\mathbb{E}\left\{\left(\sum_{i=1}^n X_i\right)^2 - 2\sum_{i=1}^n X_i^2\right\} = 0.$$

(c) Show that there does not exist a complete sufficient statistic for μ .

Solution.

(a) $f(\mathbf{x} \mid \mu) = (2\pi\mu^2)^{-n/2} \exp\left(-\frac{1}{2\mu^2} \sum x_i^2 + \frac{1}{\mu} \sum x_i - \frac{n}{2}\right)$. By Fisher–Neyman, $(T_1, T_2) = (\sum X_i, \sum X_i^2)$ is sufficient for μ .

(b) With $\mathbb{E}[X_i] = \mu$, $\text{Var}(X_i) = \mu^2$:

$$\mathbb{E}[X_i^2] = 2\mu^2, \quad \mathbb{E}[\sum X_i^2] = 2n\mu^2, \quad \mathbb{E}[(\sum X_i)^2] = n\mu^2 + n^2\mu^2 = n(n+1)\mu^2.$$

Therefore

$$\mathbb{E}[(n+1)\sum_{i=1}^n X_i^2 - 2(\sum_{i=1}^n X_i)^2] = 2n(n+1)\mu^2 - 2n(n+1)\mu^2 = 0.$$

(Note: as printed in lecture the coefficient 2 on $\sum X_i^2$ gives $\mathbb{E}[\cdot] = n(n-3)\mu^2$, which only vanishes when $n = 3$. The general identity above is the intended one.)

(c) Apply Thm 6.1: $L(\mu \mid \mathbf{x})/L(\mu \mid \mathbf{y})$ is independent of μ iff $\sum x_i = \sum y_i$ and $\sum x_i^2 = \sum y_i^2$. Hence (T_1, T_2) is **minimal sufficient**. Any complete sufficient statistic must be minimal sufficient (Lect. 6 remark), so it is in 1-1 correspondence with (T_1, T_2) . But part (b) gives $h(T_1, T_2) = (n+1)T_2 - 2T_1^2 \not\equiv 0$ with $\mathbb{E}h = 0$, contradicting completeness. Therefore no complete sufficient statistic exists for μ .

Final 25 Q3

Problem. Consider a random sample $X_1, \dots, X_n$ from a binomial (m, p) distribution. Let $\theta = P\{X = m\}$. **(a)** Show that $T = \sum_{i=1}^n X_i$ is complete and sufficient for θ . **(b)** Show that

$$U = \begin{cases} 1, & \text{if } X = m, \\ 0, & \text{otherwise,} \end{cases}$$

is unbiased for θ . **(c)** Show that the uniform minimum variance unbiased estimator (UMVUE) of θ is

$$U^*(T) = \frac{\binom{T}{m}}{\binom{mn}{m}}, \quad T = m, m+1, \dots, mn.$$

Solution.

(a) $L(p) \propto p^{\sum x_i} (1-p)^{nm-\sum x_i}$ — exponential family of full rank in $\eta = \log \frac{p}{1-p}$ with sufficient statistic $T = \sum X_i$. By Thm 6.2, T is complete and sufficient. (Equivalently, $T \sim \text{Bin}(nm, p)$, and $\mathbb{E}h(T) = 0 \Rightarrow$ polynomial in $p/(1-p)$ vanishes $\Rightarrow h \equiv 0$.)

(b) $\mathbb{E}[U] = P(X_1 = m) = p^m = \theta$.

(c) Apply Lehmann–Scheffé: $\text{UMVUE} = \mathbb{E}[U \mid T] = P(X_1 = m \mid T)$. Conditional on $T = t$:

$$P(X_1 = m \mid T = t) = \frac{P(X_1 = m) P(\sum_{i=2}^n X_i = t - m)}{P(T = t)} = \frac{p^m \binom{(n-1)m}{t-m} p^{t-m} (1-p)^{(n-1)m-(t-m)}}{\binom{nm}{t} p^t (1-p)^{nm-t}}.$$

The p - and $(1-p)$ -powers cancel and

$$\frac{\binom{(n-1)m}{t-m}}{\binom{nm}{t}} = \frac{((n-1)m)! t!}{(t-m)! (nm)!} = \frac{t!}{m! (t-m)!} \cdot \frac{m! ((n-1)m)!}{(nm)!} = \frac{\binom{t}{m}}{\binom{nm}{m}}.$$

Hence

$$U^*(T) = \frac{\binom{T}{m}}{\binom{nm}{m}}, \quad T = m, m+1, \dots, nm.$$

Final 25 Q4

Problem. Let X be a random variable with probability mass function

$$H_0 : p(x) = p_0(x), \quad H_1 : p(x) = p_1(x),$$

where

X	1	2	3	4	5	6
$p_0(x)$	.01	.01	.01	.01	.01	.95
$p_1(x)$	.05	.04	.03	.02	.01	.85

(a) Calculate the likelihood ratio L_1/L_0 for each of the values of X .

(b) By using the Neyman–Pearson lemma and your answer in (a), obtain the rejection region for the most powerful test.

(c) Obtain the most powerful size $\alpha = 0.05$ test and calculate its power.

Solution.

(a) Likelihood ratios $L_1/L_0 = p_1(x)/p_0(x)$:

X	1	2	3	4	5	6
L_1/L_0	5	4	3	2	1	$17/19 \approx 0.8947$

(b) By Neyman–Pearson, the MP test rejects for the largest values of L_1/L_0 . Sorting the x 's by L_1/L_0 descending: 1, 2, 3, 4, 5, 6.

(c) Cumulative H_0 probability when adding outcomes in this order: .01, .02, .03, .04, .05, 1. The largest size ≤ 0.05 region is $\{1, 2, 3, 4, 5\} = \{X \neq 6\}$, with exact size 0.05. So the MP $\alpha = 0.05$ test is

$$\phi(x) = \mathbf{1}_{\{x \neq 6\}}.$$

Power: $P_{H_1}(X \neq 6) = 0.05 + 0.04 + 0.03 + 0.02 + 0.01 = 0.15$.

HW3 Q5

■ **Problem.** Show that the $\text{Uniform}(\theta, \theta + 1)$ family is not complete.

Solution. For $n \geq 2$, set $Y_i = X_i - \theta \sim \text{Uniform}(0, 1)$. The range $X_{(n)} - X_{(1)} = Y_{(n)} - Y_{(1)} \sim \text{Beta}(n - 1, 2)$ has mean $(n - 1)/(n + 1)$, free of θ . Hence

$$h(\mathbf{X}) = X_{(n)} - X_{(1)} - \frac{n-1}{n+1}$$

satisfies $\mathbb{E}_\theta h \equiv 0$ but $h \not\equiv 0$.

For $n = 1$, take $h(X) = \sin(\pi X)$: $\mathbb{E}_\theta h = \int_\theta^{\theta+1} \sin(\pi x) dx = 0$ for every θ , yet $h \not\equiv 0$. So the family is not complete.

HW3 Q6

■ **Problem.** Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bin}(m, p)$. Find the UMVUE of $p^r q^s$, where $q = 1 - p$ and r, s are known positive integers with $r + s < mn$. **(a)** Distribution of $T = \sum X_i$. **(b)** Solve $\mathbb{E}[U(T)] = p^r q^s$ for U . **(c)** Argue $U(T)$ is the UMVUE.

Solution.

(a) $T \sim \text{Bin}(mn, p)$ (sum of independent binomials).

(b) $\mathbb{E}[U(T)] = p^r q^s$ becomes

$$\sum_{t=0}^{mn} U(t) \binom{mn}{t} p^{t-r} q^{mn-t-s} = 1.$$

Match this to the identity (with $k = t - r$)

$$1 = (p + q)^{mn-r-s} = \sum_{k=0}^{mn-r-s} \binom{mn-r-s}{k} p^k q^{mn-r-s-k},$$

giving $U(t) \binom{mn}{t} = \binom{mn-r-s}{t-r}$ on $t \in \{r, \dots, mn - s\}$ and $U(t) = 0$ otherwise:

$$U(T) = \frac{\binom{mn-r-s}{T-r}}{\binom{mn}{T}}, \quad T = r, r + 1, \dots, mn - s.$$

(c) $\text{Bin}(m, p)$ is a one-parameter exponential family of full rank in $\eta = \log \frac{p}{1-p}$, so T is complete and sufficient (Thm 6.2). By Lehmann–Scheffé, $U(T)$ is the unique UMVUE.

HW4 Q1

Problem. $f(x | \theta) = \theta x^{\theta-1}$, $0 < x < 1$. With $n = 1$, find the power of the MP test of $H_0 : \theta = 1$ vs $H_1 : \theta = 2$ at $\alpha = 0.05$.

Solution. NP ratio $L(2)/L(1) = 2x$ is increasing in x , so the MP test rejects when $X > c$. Size: $P_{\theta=1}(X > c) = 1 - c = 0.05 \Rightarrow c = 0.95$. Power:

$$P_{\theta=2}(X > 0.95) = \int_{0.95}^1 2x \, dx = 1 - 0.9025 = \boxed{0.0975}.$$

HW4 Q2

Problem. Same density. With $n = 2$, show the MP rejection region of size α is $X_1 X_2 \geq k^*$, where $k^* - k^* \log k^* = 1 - \alpha$.

Solution. NP ratio $L(2)/L(1) = 4X_1 X_2$, so reject when $X_1 X_2 > k^*$. Size:

$$\alpha = P_{\theta=1}(X_1 X_2 > k^*) = \int_{k^*}^1 \int_{k^*/x_1}^1 dx_2 \, dx_1 = \int_{k^*}^1 \left(1 - \frac{k^*}{x_1}\right) dx_1 = 1 - k^* + k^* \log k^*.$$

Equivalently, $1 - \alpha = k^* - k^* \log k^*$. $\square$

HW4 Q3

Problem. $f(y | \theta) = \frac{1}{2}(1 + \theta y)$, $-1 < y < 1$, $\theta > -1$. Random sample $Y_1, \dots, Y_n$. **(a)** Form of MP region for $H_0 : \theta = 0$ vs $H_1 : \theta = 1$ at level α . **(b)** $n = 1$, $\alpha = 0.05$: dividing point. **(c)** Power at $\theta = 1$ when $n = 1$, $\alpha = 0.05$.

Solution.

(a) NP ratio $L(1)/L(0) = \prod_{i=1}^n (1 + Y_i)$, so reject when $\prod (1 + Y_i) > c$. Equivalently, with $W_i = -\log((1 + Y_i)/2)$, reject when $\sum W_i < k$. (Under H_0 , $(1 + Y_i)/2 \sim \text{Uniform}(0, 1)$, so $W_i \sim \text{Exp}(1)$ and $\sum_{i=1}^n W_i \sim \text{Gamma}(n, 1)$; the threshold k is the α -quantile of that distribution.)

(b) $n = 1$: reject when $1 + Y > c$, i.e., $Y > c - 1$. Under H_0 , $Y \sim \text{Uniform}(-1, 1)$, so $\alpha = (2 - c)/2 \Rightarrow c - 1 = 1 - 2\alpha = 0.9$. **Reject iff $Y > 0.9$.**

(c) Power at $\theta = 1$:

$$P_{\theta=1}(Y > 0.9) = \int_{0.9}^1 \frac{1}{2}(1 + y) \, dy = \frac{1}{2} \left[y + \frac{y^2}{2} \right]_{0.9}^1 = \boxed{0.0975}.$$

HW4 Q4

Problem. $X_i \stackrel{iid}{\sim} \text{Poisson}(\lambda)$ for $i = 1, \dots, 10$. Construct the randomized MP test of size $\alpha = 0.05$ for $H_0 : \lambda = 0.1$ vs $H_1 : \lambda > 0.1$.

Solution. Likelihood has MLR in $T = \sum_{i=1}^{10} X_i$, and $T \sim \text{Poisson}(n\lambda) = \text{Poisson}(1)$ under H_0 . By Thm 8.1, the UMP test is

$$\phi = \mathbf{1}_{\{T > c\}} + \gamma \mathbf{1}_{\{T=c\}}, \quad P_{H_0}(T > c) + \gamma P_{H_0}(T = c) = 0.05.$$

Under $\text{Poisson}(1)$: $P(T \leq 3) \approx 0.9810$, $P(T > 3) \approx 0.0190$, $P(T = 3) \approx 0.0613$. Choose $c = 3$:

$$\gamma = \frac{0.05 - 0.0190}{0.0613} \approx 0.506.$$

HW5 Q1

Problem. $p \sim \text{Uniform}(0, 1) = \text{Beta}(1, 1)$. Observe $X = 2$ defective items out of $n = 30$. Find $P(p \leq 0.1 \mid X = 2)$.

Solution. Beta-binomial conjugacy: posterior $\sim \text{Beta}(1 + 2, 1 + 30 - 2) = \text{Beta}(3, 29)$. Hence

$$P(p \leq 0.1 \mid X = 2) = \text{pbeta}(0.1, 3, 29) \approx \boxed{0.6114}.$$

HW5 Q2

Problem. $X \mid \theta \sim N(\theta, 100)$, prior $\theta \sim N(100, 225)$, observed $x = 115$. **(a)** Show the posterior is $N(110.38, 69.23)$. **(b)** Show the Bayes factor for $H_0 : \theta \leq 100$ vs $H_1 : \theta > 100$ is 0.12.

Solution.

(a) Normal-normal conjugacy (Ex. 10.1). Posterior precision and mean:

$$B^2 = \left(\frac{1}{100} + \frac{1}{225} \right)^{-1} \approx 69.23, \quad A = B^2 \left(\frac{115}{100} + \frac{100}{225} \right) \approx 110.38.$$

(b) Posterior tail: $P(\theta \leq 100 \mid x) = \Phi((100 - 110.38)/\sqrt{69.23}) = \Phi(-1.247) \approx 0.106$. Prior is symmetric about 100, so prior odds equal 1, and

$$BF_{01} = \frac{0.106/0.894}{1} \approx \boxed{0.12}.$$

HW5 Q3

Problem. $X \sim \text{Bin}(20, p)$, observe $X = 15$. **(a)** Posterior Bayes estimate under $\text{Beta}(3, 5)$ prior. **(b)** MLE. **(c)** Repeat (a) under $\text{Uniform}(0, 1)$ prior. **(d)** Compare all three when $X = 150$ out of 200.

Solution.

(a) Posterior $\sim \text{Beta}(3 + 15, 5 + 5) = \text{Beta}(18, 10)$; mean $= 18/28 \approx 0.64$.

(b) $\hat{p}_{\text{MLE}} = 15/20 = 0.75$.

(c) Posterior $\sim \text{Beta}(16, 6)$; mean $= 16/22 \approx 0.73$.

(d) $\hat{p}_{\text{MLE}} = 150/200 = 0.75$. $\text{Beta}(3, 5)$ posterior mean $= 153/208 \approx 0.74$. $\text{Beta}(1, 1)$ posterior mean $= 151/202 \approx 0.75$. With more data, prior influence shrinks; both Bayes estimates converge to the MLE.

HW5 Q4

Problem. $X \mid \theta \sim \text{Gamma}(\alpha, 1/\theta)$, $\theta \sim \text{Gamma}(\beta, 1/\gamma)$. Find the marginal density of X .

Solution. Marginalize over θ :

$$f(x) = \int_0^\infty \frac{\theta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\theta x} \cdot \frac{\gamma^\beta}{\Gamma(\beta)} \theta^{\beta-1} e^{-\gamma\theta} d\theta = \frac{x^{\alpha-1} \gamma^\beta}{\Gamma(\alpha)\Gamma(\beta)} \int_0^\infty \theta^{\alpha+\beta-1} e^{-\theta(x+\gamma)} d\theta.$$

The inner integral equals $\Gamma(\alpha + \beta)/(x + \gamma)^{\alpha+\beta}$, so

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\gamma^\beta x^{\alpha-1}}{(x + \gamma)^{\alpha+\beta}} \quad (x > 0).$$

(A generalized beta-prime / compound gamma density.)

HW5 Q5

Problem. Pareto prior $\pi(\theta) = \gamma\beta^\gamma\theta^{-(\gamma+1)}$, $\theta \geq \beta$, with $\beta > 0$, $\gamma > 2$. **(a)** Prior mean. **(b)** Given $Y_1, \dots, Y_n \stackrel{iid}{\sim} \text{Uniform}(0, \theta)$, posterior mean. **(c)** Show $\hat{\theta} = \frac{\gamma + n}{\gamma + n - 1} \max\{z, \beta\}$, $z = \max(y_1, \dots, y_n)$, is admissible under squared-error loss.

Solution.

(a) $\mathbb{E}\theta = \int_\beta^\infty \theta \cdot \gamma\beta^\gamma\theta^{-(\gamma+1)} d\theta = \gamma\beta^\gamma \cdot \frac{\beta^{1-\gamma}}{\gamma-1} = \frac{\gamma\beta}{\gamma-1}.$

(b) Likelihood $\propto \theta^{-n} \mathbf{1}_{\{\theta \geq \max y_i\}}$. With $z_* = \max\{y_1, \dots, y_n, \beta\}$, the posterior kernel is $\theta^{-(n+\gamma+1)} \mathbf{1}_{\{\theta > z_*\}}$, normalized to

$$\pi(\theta \mid \mathbf{y}) = (n + \gamma) z_*^{n+\gamma} \theta^{-(n+\gamma+1)} \mathbf{1}_{\{\theta > z_*\}} \quad (\text{Pareto on } (z_*, \infty)).$$

Posterior mean:

$$\mathbb{E}[\theta \mid \mathbf{y}] = \frac{n + \gamma}{n + \gamma - 1} z_*.$$

(c) This posterior mean equals $\hat{\theta}$. Under squared-error loss, the posterior mean is the unique Bayes estimator (Cor. 10.1); a unique Bayes estimator with finite Bayes risk is admissible.

HW5 Q6

Problem. Crash probabilities: plane 5×10^{-6} , car 10^{-2} . Loss table:

	Plane	Car
Crash	10^6	100
No crash	5	0

Which decision has smaller risk?

Solution. Expected loss:

$$R_{\text{plane}} = 10^6(5 \times 10^{-6}) + 5(1 - 5 \times 10^{-6}) \approx 5 + 5 = 10,$$

$$R_{\text{car}} = 100(10^{-2}) + 0(1 - 10^{-2}) = 1.$$

Travel by car is the decision with smaller risk.

HW5 Q7

Problem. Risk table with prior $\{1/8, 3/8, 1/4, 1/4\}$ on $\{\theta_1, \theta_2, \theta_3, \theta_4\}$:

	d_1	d_2	d_3
θ_1	0	2	3
θ_2	1	0	2
θ_3	3	4	0
θ_4	1	2	0

(a) Bayes risk of each decision. (b) Bayes decision. (c) Minimax decisions.

Solution.

(a)

$$r(d_1) = \frac{1}{8} \cdot 0 + \frac{3}{8} \cdot 1 + \frac{1}{4} \cdot 3 + \frac{1}{4} \cdot 1 = \frac{11}{8},$$

$$r(d_2) = \frac{1}{8} \cdot 2 + \frac{3}{8} \cdot 0 + \frac{1}{4} \cdot 4 + \frac{1}{4} \cdot 2 = \frac{14}{8},$$

$$r(d_3) = \frac{1}{8} \cdot 3 + \frac{3}{8} \cdot 2 + \frac{1}{4} \cdot 0 + \frac{1}{4} \cdot 0 = \frac{9}{8}.$$

(b) d_3 minimizes Bayes risk.

(c) Worst-case losses: $\max_{\theta} L(\theta, d_1) = 3$, $\max L(\cdot, d_2) = 4$, $\max L(\cdot, d_3) = 3$. Minimax decisions: d_1 and d_3 .

HW5 Q8

Problem. $X \sim \text{Bernoulli}(\theta)$. Decision rule $d(1) = d_1$, $d(0) = d_2$, with squared-error loss $L(d, \theta) = (d - \theta)^2$. (a) Risk function $R_d(\theta)$. (b) Show the minimax decision satisfies $d_1 = 3/4$, $d_2 = 1/4$.

Solution.

(a) $R_d(\theta) = \mathbb{E}[L(\theta, d(X))] = (d_1 - \theta)^2\theta + (d_2 - \theta)^2(1 - \theta)$.

(b) From Ex. 10.9, the minimax estimator of θ for $\text{Bin}(n, \theta)$ under squared-error loss is

$$T^* = \frac{X + \sqrt{n}/2}{n + \sqrt{n}} \quad (\text{Bayes vs. Beta}(\sqrt{n}/2, \sqrt{n}/2) \text{ prior, with constant risk}).$$

For $n = 1$: $T^*(X) = (X + 1/2)/2$, so $T^*(1) = 3/4$ and $T^*(0) = 1/4$. Hence $\boxed{d_1 = 3/4, d_2 = 1/4}$.